4. Coordinate systems

What students need to learn:

Cartesian equations for the parabola and rectangular hyperbola.	Students should be familiar with the equations: $y^2 = 4ax$ and $xy = c^2$
Idea of parametric equations for parabola and rectangular hyperbola.	$x = at^2$, $y = 2at$ and $x = ct$, $y = \frac{c}{t}$. The idea of $(at^2, 2at)$ and $(ct, \frac{c}{t})$ as general points on the parabola and rectangular hyperbola respectively.
The focus-directrix property of the parabola.	Concept of focus and directrix and parabola as locus of points equidistant from focus and directrix.
Tangents and normals to these curves.	Differentiation of $y = 2a^{\frac{1}{2}}x^{\frac{1}{2}}$, $\quad y = \frac{c^2}{x}$. Parametric differentiation is not required.